

SERVICE MANUAL

MRI Devices Corporation
3545 SW 47th Avenue
Gainesville, FL 32608

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GE Signa® 1.5T and/or 1.0T Open Neurovascular Array

GE Catalog Part Number:	M1087NB	(1.5T)
	M1887NB	(1.0T)
MRIDC Part Number:	800202	(1.5T)
	800201	(1.0T)



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Rev. 11/15/2000

Language Policy For Service Documentation (Dir. 2128126)

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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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SECTION 1 - INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE Signa 1.5Tand/or 1.0T Open Neurovascular Array.



1.5T SHIPPING LIST – TABLE 1-1-1

Description	GE Part #	MRIDC Part #	Qty
Neurovascular Assembly with Cable	2293668-2	101413	1
Neurovascular Operator Manual	2293668-3	500067	1
Neurovascular Service Manual	2293668-4	TR0112S	1
Phantom Positioner	2293668-7	101655	1
Wedge Pad	E8800NB	101407	2
Pad	E8800NC	101408	1

1.0T SHIPPING LIST – TABLE 1-1-2

Description	GE Part #	MRIDC Part #	Qty
Neurovascular Assembly with Cable	2293669-2	101423	1
Neurovascular Operator Manual	2293668-3	500067	1
Neurovascular Service Manual	2293668-4	TR0112S	1
Phantom Positioner	2293668-7	101655	1
Wedge Pad	E8800NB	101407	2
Pad	E8800NC	101408	1

1-2 Compatibility

This coil is compatible with the following hardware configurations:

- Signa 1.5T LX/Horizon and CVNV
- Signa 1.0T LX/Horizon and CVNV

1-3 Related Documentation

- Signa LX Service Methods CD, 2160623-1
- 1.5T/1.0T Open Neurovascular Array Operator Manual, 2293668-3

1-4 Environmental Requirements

Storage Requirements

The Coil should be stored in the Scanner Room.

Dimensions

1.5T Coil Dimension:	679.5 mm x 394.0 mm x 415.9 mm	(26.75 in x 15.50 in x 16.375 in)
1.0T Coil Dimension:	679.5 mm x 394.0 mm x 415.9 mm	(26.75 in x 15.50 in x 16.375 in)
Phantom Holder	453.6 mm x 381.0 mm x 142.2 mm	(17.86 in x 15.00 in x 5.60 in)

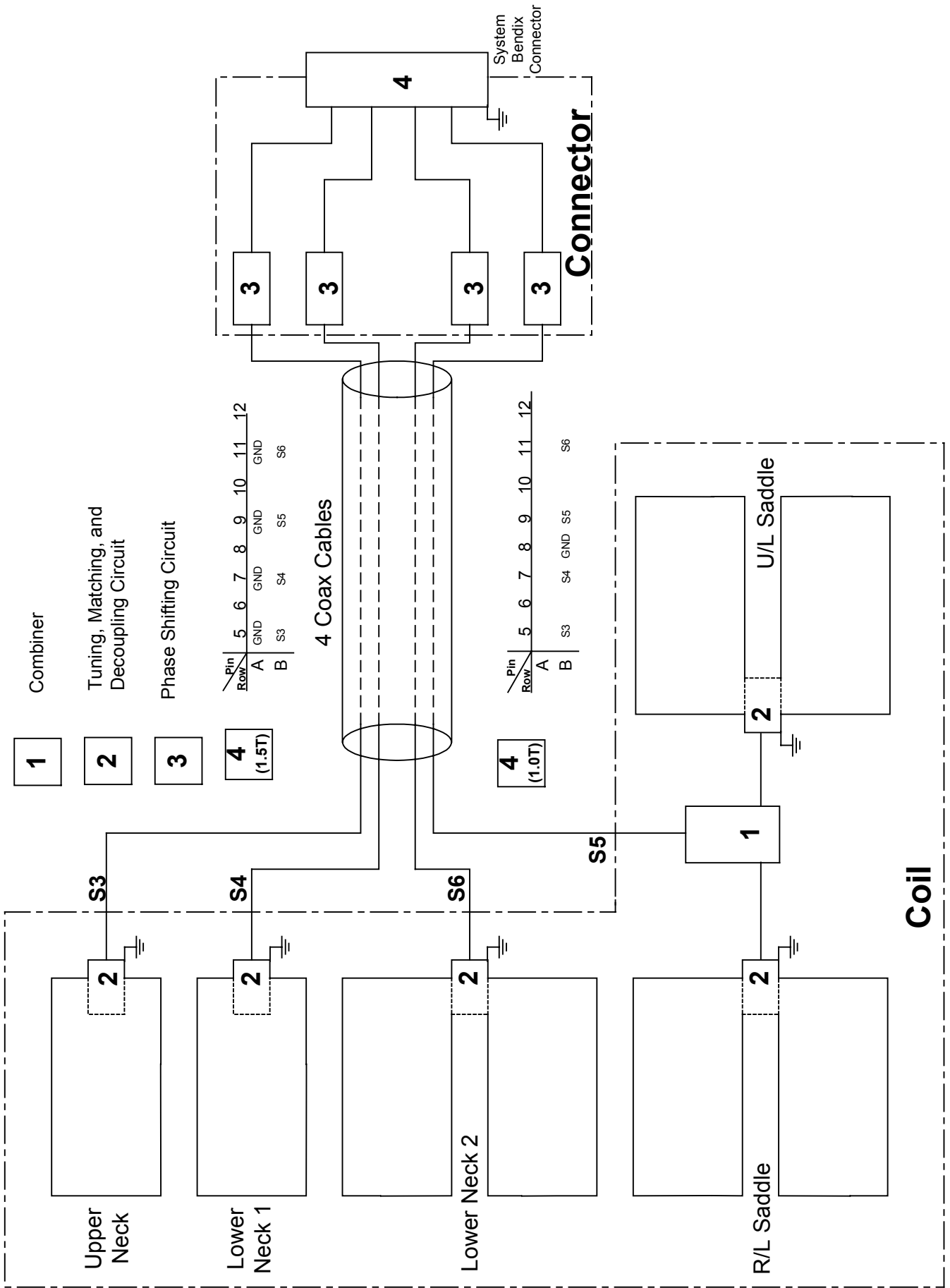
Weight

1.5T Coil Weight:	9.77 Kg	(21.5 lb.)
1.5T Coil Weight:	9.77 Kg	(21.5 lb.)

1-5 Theory of Operation

Refer to the block diagram on the following page.

The Phased Array Neurovascular coil consists of a top and bottom piece which must be securely fastened together for safe effective operation. The coil utilizes four channels of the multi-coil system. The head channel on S6 is a quadrature saddle coil which uses the combiner labeled (1). The neck coverage comes from three channels, one for the anterior neck and the other two for the posterior neck. The coils are tuned/matched and decoupled through the circuits labeled (2). Decoupling is actively controlled using the system bias current and trap circuits which produce high impedance in the coil loops when PIN diodes are forward biased. If bias current is not available due to a system failure, the traps will be activated passively by the RF power during transmission. These trap circuits ensure safe artifact free operation. Phase shift circuits (labeled {3}) are employed to improve inductive isolation using the system preamplifier impedance mismatch. Item 4 is the system interface which defines the pins associated with each coil channel.



SECTION 2 - SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None

2-1-2 Installing the Coil

The names for this coil are: NVHEAD, NVARRAY, and NVFASTHD.

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

If the coil does not exist in Coil Config File refer to the Adding New Coils to Config File Manager procedure and use the coil configuration information in Section 7-3 of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3 - Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coils is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3 - Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	Supplier Part #	Qty
18cm diameter spherical TLT Phantom	46-265826G6	N/A	1
10cm diameter spherical phantom	46-317586G1	N/A	1
Phantom Positioner	2293668-7	101655	1

3-2-2 Explanation of Procedure

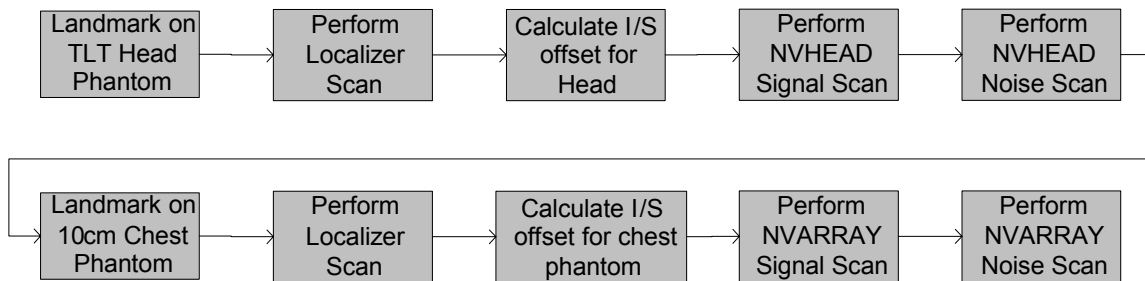
For evaluation of coil performance, images will be acquired in the axial plane. The following protocol shall be used for coil comparison and image quality analysis.

The NV-Array has two sections controlled by the coil configuration files: NVHEAD, and NVARRAY. The protocol given below shall be run using NVHEAD and NVARRAY on the 2 phantoms. The final report will include the SNR analyses on axial images.

Individual element performance: The NV-Array coil contains four elements, each of those corresponding to its own receiver. Using the protocol described here, signal should appear on all four receivers. The user may check individual element performance using Manual Prescan. When the Manual Prescan window appears, click on *Scan T/R*. In the graph at the right, each receiver should show the projection of the phantom along the phase direction for its corresponding element.

Use the following procedures to acquire the sagittal localizer, axial signal, and axial noise images. Complete this procedure TWICE: once for NVHEAD mode and once for NVARRAY mode. The procedures for NVHEAD and NVARRAY are nearly identical, with any differences noted in the procedures and protocols listed below. Perform the procedures in the order shown in Procedure Flowchart – Table 3-2-2.

PROCEDURE FLOWCHART – TABLE 3-2-2



3-2-3 Localizer Scan

The following procedure is specific to the LX platform but can be easily adapted for 5.x systems. Perform a sagittal localizer scan in order to determine the center of the phantom.

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open the protocols window.
2. Remove any other surface coils from the cradle. Referring to Figure 1, place the Neurovascular coil on the patient table with the cable oriented into the magnet. Lock the base into the slots in the table. Plug the phased array plug into the phased array port on the Signa system.

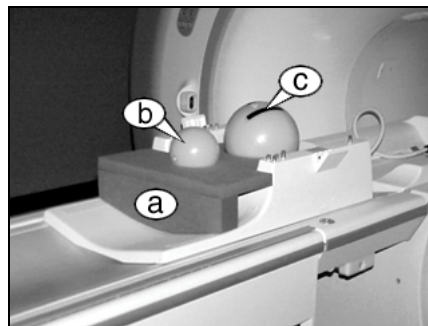
Unlatch and remove the coil top, placing it in a safe location. Place the phantom positioner into the coil as seen in Figure 2a. Then place the 10cm spherical phantom and the 18cm spherical TLT phantom onto the phantom positioner as seen in Figure 2b and 2c, respectively. Replace and latch the coil top.

Figure 1



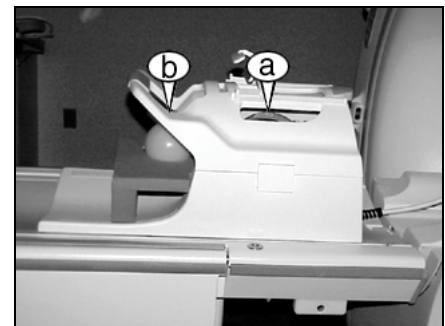
Coil Orientation

Figure 2



Phantoms and Positioner

Figure 3



Landmark

3. At the magnet, press “**Alignment Light**” button to turn on the light. Move the cradle to align the coil to the alignment lights as follows:
 - For the first localizer scan, landmark in the center of the Head TLT phantom, as indicated in Figure 3a. [use NVHEAD mode]
 - For the second localizer scan, landmark in the center of the 10cm phantom, as indicated in Figure 3b. [use NVARRAY mode]
4. Press the “**Landmark**” button to landmark the alignment.

5. Move the coil to scan position by pushing the **“Move to Scan”** button, ensuring the cable does not get snagged.
6. At the console, set the protocols using the values in the Localizer protocol column of Table 3-2-6.
7. Select **[Save Series]** then **[Prepare to Scan]**. Select **[Auto Prescan]**. When auto prescan is complete, select **[Scan]**.

3-2-4 Determining The I/S Center Location of the Phantom

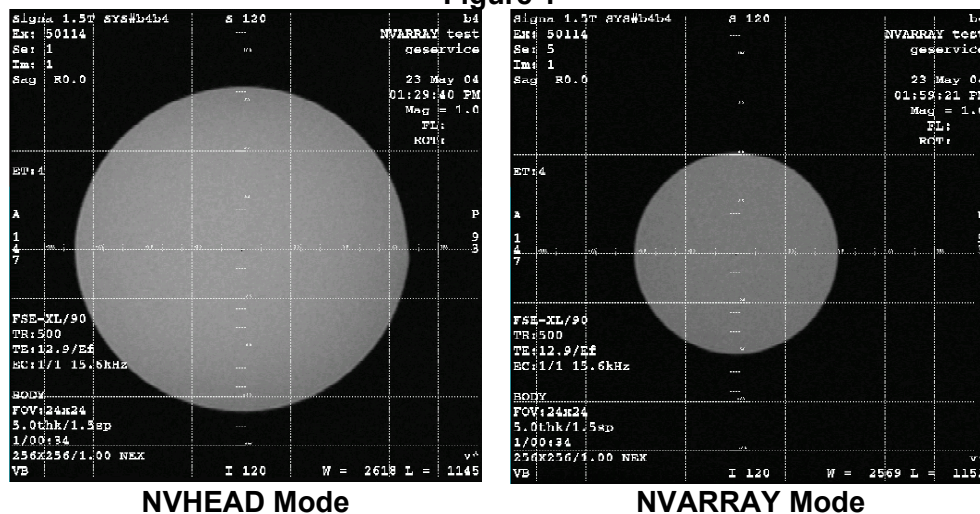
In the image browser, set the grid settings as listed in Table 3-2-4, using [User Prefs] -> [Grid prefs].

GRID SETTINGS - TABLE 3-2-4

Matrix Lines	On
Line Style	Dotted
Grid Spacing	50mm
Tick Spacing	10mm
Tick Length	5mm

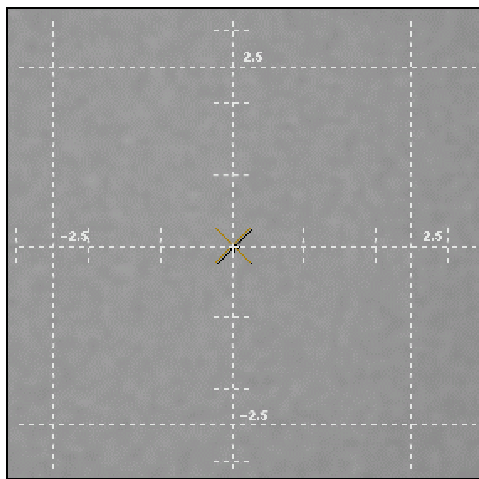
Turn on the grid. Left click on the center of the grid and move until centered on the phantom image. (Window level if necessary to make the grid visible) See Figure 1 for an example of a centered grid.

Figure 1



Click the user interface button [Measure], select the “X”, and adjust its location until the “X” is centered on the grid. See Figure 2.

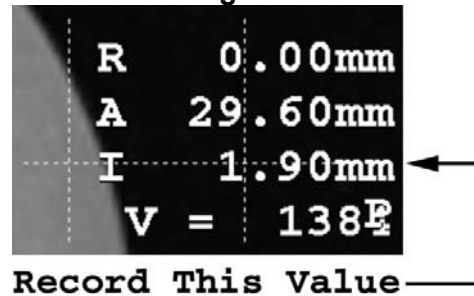
Figure 2



Center “X” on Grid

Record the S/I location of the “X” as “S/I Offset” in the SNR worksheet. The S/I location is listed on the right side of the image display, as shown in Figure 3. This value will be used in the signal and noise scans to choose the slice location.

Figure 3



3-2-5 Signal Scan

Perform the signal scan using the values in the Signal protocol column of Table 3-2-6 and the S/I location recorded in the localizer scan procedure.

1. Select [Save Series] then [Prepare to Scan].
2. Prior to the scan, open the [Display CVs] menu under Research Operations. To obtain signal images for individual receivers, set “saveinter” CV to 1.
3. Select [Auto Prescan]. Record the R1 R2 and TG values obtained, then select [Scan].

3-2-6 Noise Scan

Perform the noise scan using the values in the Noise protocol column of Table 3-2-6. Do not change the prescan parameters before running this scan.

1. Select [Save Series] then [Prepare to Scan].
2. Prior to the scan, open the [Display CVs] menu under Research Operations. To obtain the noise images, set the “saveinter”, “do_noise”, and “rh_format” CVs to 1.
3. Select [Manual Prescan]. Verify that R1, R2, TG, and Bandwidth have not changed from the previously completed signal scan. Select [Done] and exit without changing any parameters. Select [Scan].

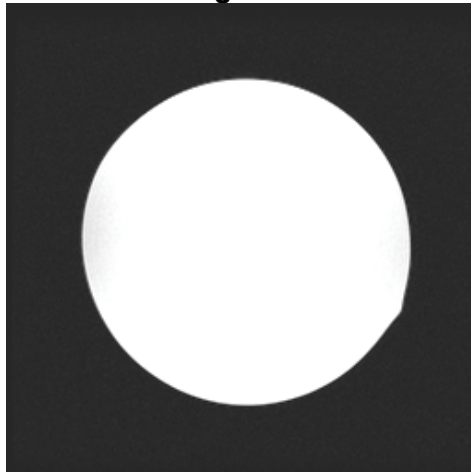
LOCALIZER, SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-6

Protocol	Localizer	Signal	Noise
<i>Patient/Exam Information</i>			
Patient ID	geservice	geservice	geservice
Patient Name	NVARRAY QA	NVARRAY QA	NVARRAY QA
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)	111 lbs. (50 kg)
<i>Patient Position</i>			
Patient Position	Supine	Supine	Supine
Patient Entry	Head First	Head First	Head First
Coil First Scan	Body	NVHEAD	NVHEAD
Coil Second Scan	Body	NVARRAY	NVARRAY
First Series Description	sag loc nvhead	ax sig nvhead	ax Noise NVHEAD
Second Series Description	sag loc nvarray	ax sig nvarray	ax Noise NVARRAY
<i>Imaging Parameters</i>			
Plane	Sagittal	Axial	Axial
Mode	2D	2D	2D
Pulse Seq	FSE-XL	FSE-XL	GRE
Imaging Options	Fast	Fast	None
PSD Name	<i>leave blank</i>	<i>leave blank</i>	<i>leave blank</i>
Protocol	<i>leave blank</i>	<i>leave blank</i>	<i>leave blank</i>
<i>Scan Timing</i>			
# of Echoes	1	1	1
TE	17	17	Minfull
TR	500	500	34
Echo Train Length/Flip Angle	4	4	1
Bandwidth	15.63	15.63	15.63
<i>Additional Parameters</i>			
<i>No Entries Required ►</i>			
<i>Acquisition Timing</i>			
Freq	256	256	256
Phase	256	256	256
NEX	1	1	1
Phase FOV	1.00	1.00	1.00
Freq DIR	S/I	A/P	A/P
Auto Center Freq	Peak	Peak	Peak
Autoshim	On	On	On
Phase Correct	On	On	N/A
Contrast	Off	Off	Off
# of Reps B4 Pause	0	0	0
<i>Scanning Range</i>			
FOV	24	24	24
Slice Thickness	5	3.0	3.0
Spacing	1.5	1.5	1.5
Start R/L	0.0		
Start S/I		[S/I offset]	[S/I offset]
P/A Center	A27.0	A27.0	A27.0
S/I Center	0.0		
L/R Center		0.0	0.0
End R/L	0.0		
End S/I		[S/I offset]	[S/I offset]
Slices	1	1	1
Table Delta	0.00	0.00	0.00

3-2-7 SNR Image Analysis

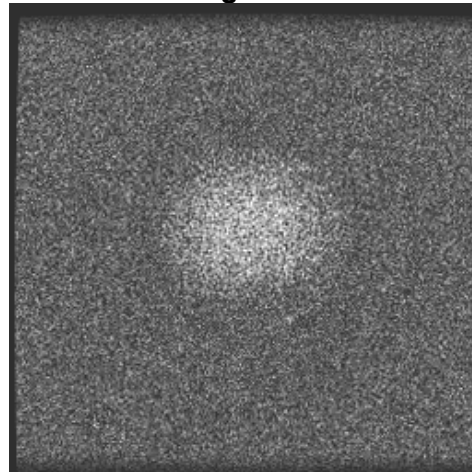
Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button *Measure*, select the circular or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image. Examples of typical Receiver Images are shown in Figures 1 through 6 below. These images represent the order of images on an LX system. **Note that images on an Excite system will appear in a different order, as detailed on the SNR worksheet.**

Figure 1



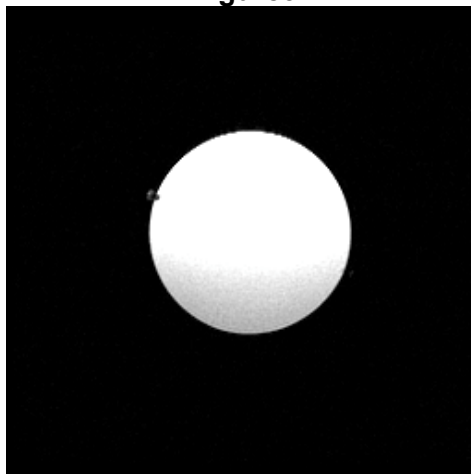
NVHEAD Receiver 1

Figure2



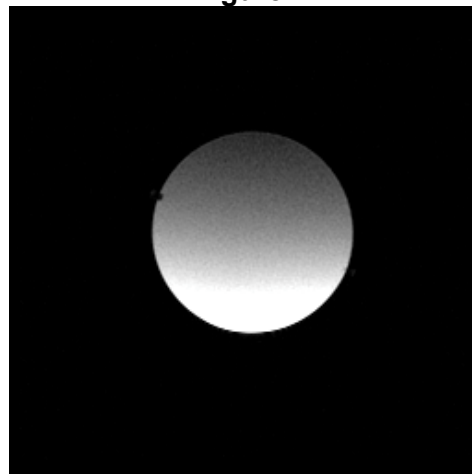
NVARRAY Receiver 1
Not Used for SNR Analysis

Figure3

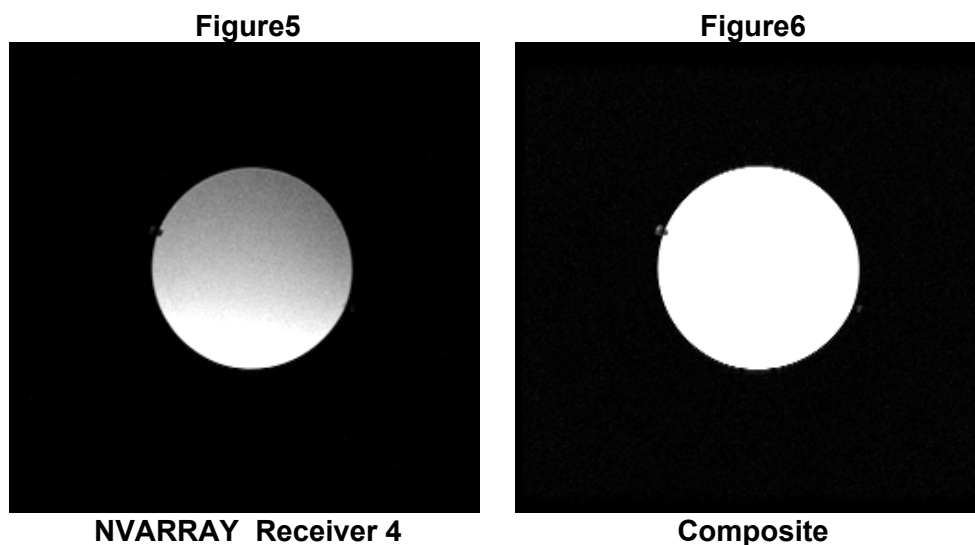


NVARRAY Receiver 2

Figure4



NVARRAY Receiver 3



3-2-8 SNR Measurement

For the signal measurement, choose a circular ROI covering the appropriate section of the phantom for the receiver channel being scanned. For each signal series:

Center a grid on the phantom image. Using the **[Measure]** tool, set a circular ROI with its center at the center of the grid. Enlarge the ROI to the following size:

NVHEAD	15cm diameter
NVARARRAY	8cm diameter

Record the mean signal for each receiver in the SNR worksheet.

For each noise image (1 from the NVHEAD receiver and 3 from the active NVARRAY receivers, measure the noise with a 20cm square. Record the standard deviation of the noise for each receiver in the SNR worksheet.

The SNR shall be calculated using the signal to noise ratios of the individual receiver channels. Individual receiver SNR is defined as the mean of data within the signal ROI divided by the standard deviation of data within the noise ROI:

$$SNR_i = \frac{\text{mean of signal within ROI}}{\text{standard deviation of noise within ROI}}$$

3-2-9 Decision Criteria

Overall SNR performance:

The SNR measurements shall meet the following lower spec limits:

NVHEAD	135
NVARARRAY Receiver 1	61
NVARARRAY Receiver 2	44
NVARARRAY Receiver 3	44

3-3 External Cable Check

Not Applicable

3-4 PIN Diodes Check

A digital multimeter may be used to measure the PIN diode drop across any of the elements. A single PIN diode drop of 0.7 ± 0.1 volt should be seen at S3, S4 and S6, with the black lead to ground. Reversing the leads, a reading of 2.6 ± 0.2 volts should be seen.

S5* is connected to two saddle elements through a combiner, as shown in the Block Diagram in Section 1-5. Therefore, the S5* readings differ as shown in Table 3-4.

If either reading is a short circuit, the diode(s) or the cable is shorted, and a replacement cable should be used repeat this check. The scanner will typically detect open or shorted channels and display a scan error.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Positive Lead	Negative Lead	Reading
S3	GND	0.7 ± 0.1 Volt
S4	GND	0.7 ± 0.1 Volt
S5*	GND	0.6 ± 0.1 Volt
S6	GND	0.7 ± 0.1 Volt
GND	S3	2.6 ± 0.2 Volts
GND	S4	2.6 ± 0.2 Volts
GND	S5*	2.0 ± 0.2 Volts
GND	S6	2.6 ± 0.2 Volts

3-5 Mechanical Hardware Check

Not Applicable

3-6 Troubleshooting Tips

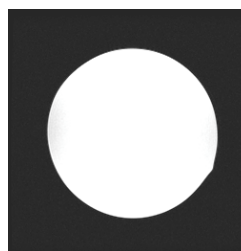
If poor image quality or dead channels are present, use the following setup information and compare the resulting images to isolate any defective part(s).

NVHEAD TROUBLESHOOTING – TABLE 3-6-1

Element Name	Receiver 1	Receiver 2	Receiver 3	Receiver 4	not used	not used	not used	not used	not used
channel number	1	2	3	4	5	6	7	8	
selected channel	See Below				no	no	no	no	
mc bias driver #	1	2	3	4	5	6	7	8	
active bias driver	yes	yes	yes	yes	no	no	no	no	

selected channel	yes	no	no	no
---------------------	-----	----	----	----

Receiver 1 ►



NVARRAY TROUBLESHOOTING – TABLE 3-6-2

Element Name	Receiver 1	Receiver 2	Receiver 3	Receiver 4	not used	not used	not used	not used	not used
channel number	1	2	3	4	5	6	7	8	
selected channel	See Below				no	no	no	no	
mc bias driver #	1	2	3	4	5	6	7	8	
active bias driver	yes	yes	yes	yes	no	no	no	no	

selected channel	yes	no	no	no
------------------	-----	----	----	----

NOTE: Not used for SNR analysis.
Image of the 10cm phantom using receiver 1 (head).

Receiver 1 ►

selected channel	no	yes	no	no
------------------	----	-----	----	----

Receiver 2 ►

selected channel	no	no	yes	no
------------------	----	----	-----	----

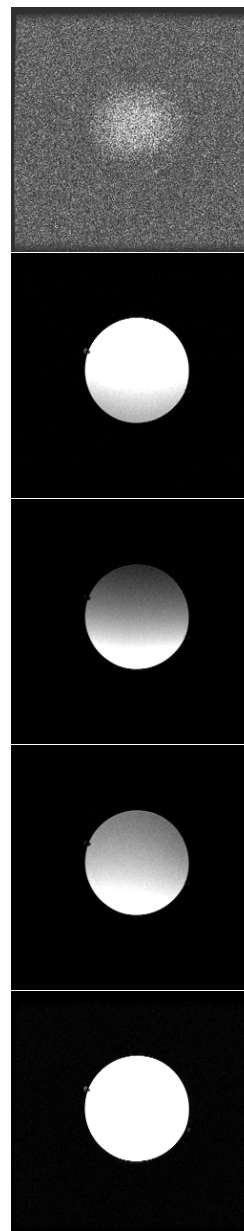
Receiver 3 ►

selected channel	no	no	no	yes
------------------	----	----	----	-----

Receiver 4 ►

selected channel	yes	yes	yes	yes
------------------	-----	-----	-----	-----

Composite ►



SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The Open Neurovascular Array pad may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

Not Applicable

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

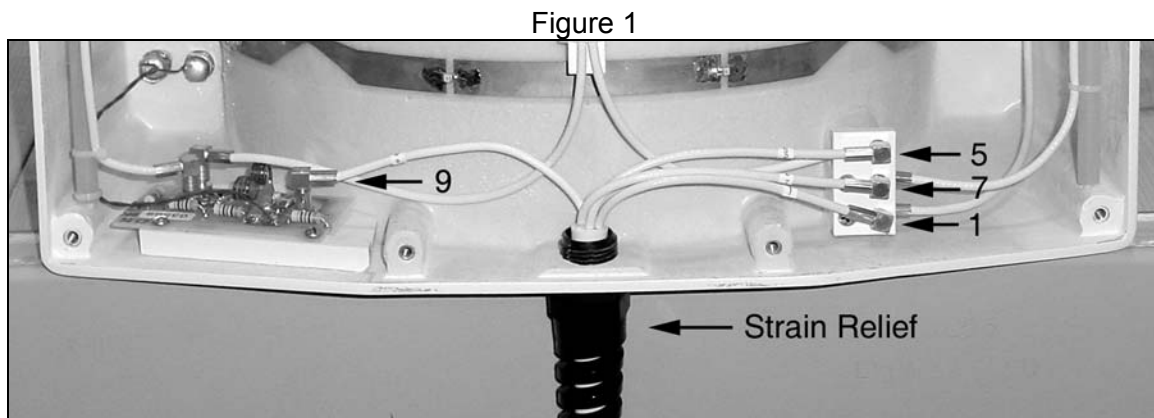
5-1 Disassembly of Coil

Not Applicable

5-2 External Cable Replacement

To remove the external cable [for part number see Table 6-1 Field Replaceable Units]:

1. Remove the 12 Phillips head screws located around the perimeter of the coil bottom. The coil bottom cover will now lift out of place.
2. Remove the bottom cover and place it in a safe location. Figure 1 is the bottom view of the coil with the cover removed.



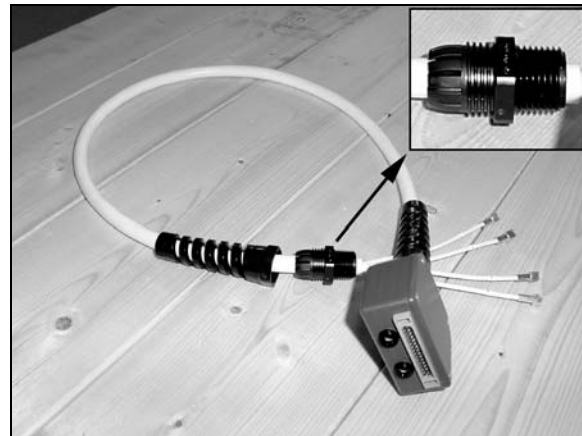
The coil cable assembly contains 4 individual coaxial cables as seen in Figure 1. Each individual coaxial cable is tagged for identification [1,5,7,9].

3. Disconnect each individual coaxial cable by pulling the connector out and away from the connector block or circuit board.
4. Group the individual coaxial cables and, using an adjustable open end wrench, loosen the strain relief plastic nut closest to the coil cover [counterclockwise] and remove the coil cable assembly.

Figure 2

The Coil Cable Assembly uses a threaded plastic piece [Figure 2 Inset] that attaches the cable assembly to the coil cover **and** tightens the strain relief around the cable assembly.

The threads on the right, and the plastic nut, attach the cable assembly to the cover. The threads on the left tighten the strain relief around the cable assembly when the strain relief nut is tightened.



Coil Cable Assembly

The steps for re-assembly using the replacement coil cable are as Follows:

1. One by one, thread the individual coaxial cables through the threaded hole in the coil cover.
2. Grouping the individual coaxial cables, screw the coil cable assembly plastic connector into the threaded hole in the coil cover. Snug the plastic nut down using an open end adjustable wrench.
3. Loosen the strain relief nut enough to allow the coil cable to rotate in the strain relief. Orient the coil cable such that, when the coil is laying with the bottom cover [and 12 screws] facing up, the system connector cover [and 4 screws] is also facing up. Snug down the strain relief nut using an open end adjustable wrench.
4. Connect the 4 individual coaxial cables using the cable tags and identifiers in Figure 1.
5. Using a Phillips head screw driver, replace the coil cover and 12 Philips head screws.

Verify coil operation according to Section 3-2, Coil Imaging Performance Verification.

5-3 Mechanical Hardware Replacement

Not Applicable

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

1.5T FIELD REPLACEABLE UNITS LIST – TABLE 6-1-1

Description	GE Part #	MRIDC Part #
1.5T Coil with Cable	2293668-2	101413
1.5T Cable only	2225476-3	100014
Latches	2225476-6	900214
Phantom Positioner	2293668-7	101655

1.0T FIELD REPLACEABLE UNITS LIST – TABLE 6-1-2

Description	GE Part #	MRIDC Part #
1.0T Coil with Cable	2293669-2	101423
1.0T Cable only	2225477-3	100015
Latches	2225476-6	900214
Phantom Positioner	2293668-7	101655

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	MRIDC Part #
Wedge Pad	E8800NB	101407
Pad	E8800NC	101408

SECTION 7 – APPENDIX

7-1 SNR Worksheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

4CH NV ARRAY SNR WORKSHEET

Coil Serial Number
System ID
Date

NVHEAD

S/I Offset
Mean Signal:

Std. Dev. Noise:

SNR:
(mean signal /
std. dev. Noise)

NVARRAY

I/S Offset

Receiver 1: LX: IMAGE 2, EXCITE: IMAGE 4

Mean Signal:

Std. Dev. Noise:

SNR:
(mean signal /
std. dev. Noise)

Receiver 2: LX: IMAGE 3, EXCITE: IMAGE 2

Mean Signal:

Std. Dev. Noise:

SNR:
(mean signal /
std. dev. Noise)

Receiver 3: LX: IMAGE 4, EXCITE: IMAGE 1

Mean Signal:

Std. Dev. Noise:

SNR:
(mean signal /
std. dev. Noise)

7-2 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation

7-3 Coil Configuration

Parameter	1.5T	1.5T	1.5T	1.0T	1.0T
Coil Name	NVHEAD	NVARRAY	NVFASTHD	NVHEAD	NVARRAY
Coil Type	3	3	3	3	3
Extremity Coil	no	no	no	no	no
Cable Loss	1.05	1.05	1.05	1.24	1.24
Coil Loss (BRM/CRM)	1.72/0.952	1.72/0.952	1.72/0.952	0.549	0.549
Recon Scale Factor	1	1	1	1	1
Linear vrs Quadrature	1	1	1	1	1
Multiple Receiver Coil?	Yes	yes	yes	yes	yes
Number of Receivers	1	4	1	1	4
Starting Receiver ID	0	0	0	0	0
Ending Receiver ID	0	3	0	0	3
Multi-Coil Port Enable	4	6	4	4	6
Multi-Coil Part Error Enable	4	6	4	4	6
Additional Transmit Attenuation	0	0	0	0	0
Number of Fast Receivers	1	0	1	0	0
Starting Fast Receiver ID	4	4	4	4	4
Ending Fast Receiver ID	4	4	4	4	4
Start TA Value	90	90	90	90	90
Start RG Value	12	12	12	12	12
Multi-Coil Recon Enable	0	0	0	0	0
Head Default Freq. Direction	1	1	1	1	1
SCIC Axial	0	0	0	0	0
SCIC Saggital	0	0	0	0	0
SCIC Coronal	0	0	0	0	0
Auto Shim Receiver	-1	-1	-1	-1	-1

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change	DCN
01	03-15-01	L. Hyler	Initial Issue	0149
1A	10-08-01	L. Hyler	Add SNR Sections	
1B	10-08-01	L. Hyler	Add MCAT Numbers	
1C	10-10-01	L. Hyler	Add Troubleshooting Tables & Final Part Numbers	
1D	10-24-01	L. Hyler	Change 1-2 Compatibility	
02	12-04-01	L. Hyler	Issue Revision 02	
03	06-14-02	L. Hyler	Correct Cable Pinout for 1.0T in Block Diagram and Correct 3-4 Pin Diodes Check Values	
04	01-05-05	L. Hyler	Implement Updated Verification and Validation Procedure [M1087NB & M1887NB Rev C]	